

Lecture 08 - Local Illumination

This lecture explains how visible surface points receive color from local light, material, normal, and view information. It is about shading a point, not solving all global light transport.

Source chunk	course_text_parts/03_lectures/08_local-illumination.txt
Extracted slide pages	62
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Slide walkthrough

Each row gives a short reading comment for one extracted slide page. Use it to turn sparse slide bullets into a human-readable path through the lecture.

Page	Slide title	How to read it
1	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
2	Local Illumination	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
3	8.1 Physics of Light	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
4	8.1 Physics of Light	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
5	Electromagnetic Radiation	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
6	Visible Light	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
7	Visible Light	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
8	Light Spectra	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.

Page	Slide title	How to read it
9	Black-Body Radiation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
10	Black-Body Radiation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
11	Spectral Density Function (SDF)	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
12	Light Perception	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
13	Interpreting SDFs	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
14	Representing SDFs	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
15	8.2 Light Sources	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
16	Light Sources	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
17	Point Light Sources	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
18	Cone Light Sources	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
19	Light Attenuation	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
20	Area Light Sources	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
21	Directed Light	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
22	Local vs. Global Illumination	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
23	Ambient Light	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.

Page	Slide title	How to read it
24	8.3 Material Models	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
25	Material Models	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
26	Materials vs. Textures	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
27	Light Material Interaction 1/3	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
28	Light Material Interaction 2/3	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
29	Light Material Interaction 3/3	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
30	Reflection = SDF x SRF	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
31	Reflection Modeling 1/2	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
32	Reflection Modeling 2/2	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
33	8.4 Phong Illumination Model	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
34	Empirical Observations	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
35	Illumination Models	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
36	yaR	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
37	Diffuse Intensity 2/3	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
38	Diffuse Intensity 3/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?

Page	Slide title	How to read it
39	Ambient Intensity 1/2	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
40	Ambient Intensity 2/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
41	Specular Intensity 1/3	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
42	Reflection Vector ■■	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
43	Specular Intensity 2/3	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
44	Specular Exponent ■	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
45	Specular Intensity 3/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
46	Blinn-Phong Illumination Model 1/2	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
47	Blinn-Phong Illumination Model 2/2	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
48	Phong Illumination Model	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
49	Phong Illumination Model	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
50	8.5 Shading	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
51	Shading	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
52	Shading Computation	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
53	Flat Shading	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.

Page	Slide title	How to read it
54	Gouraud Shading 1/5	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
55	Gouraud Shading 2/5	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
56	Gouraud Shading 3/5	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
57	Gouraud Shading 4/5	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
58	Gouraud Shading 5/5	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
59	Phong Shading 1/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
60	Phong Shading 2/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
61	Literature and other sources used in this chapter	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
62	Lesen und Ausprobieren	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.

08.1 Physics of Light

Source location: Section 8.1

Commentary

The physics slides provide intuition for light as energy interacting with surfaces. For this course, the goal is not full physical simulation but a usable model for real-time rendering.

Important terms are reflection, absorption, wavelength/color, intensity, and direction. These terms become shader inputs or material parameters later. The simplified model must still preserve the relation between light direction, surface orientation, and observed brightness.

Mental model

Shading is an approximation of light-surface interaction that is cheap enough for rendering.

Check yourself

Why does surface orientation affect diffuse brightness?

08.2 Light Sources

Source location: Section 8.2

Commentary

Light source models define where light comes from and how its direction and intensity are computed. Directional lights approximate distant sources with parallel rays. Point lights emit from a position and often include attenuation. Spotlights add a directional cone.

In shader code, the type of light determines how you compute the light vector. A directional light can use a fixed direction; a point light requires subtracting the surface position from the light position.

Mental model

Different light types mainly change how the light direction and intensity are computed at the surface point.

Check yourself

What is the difference between a directional light vector and a point light vector?

08.3 Material Models

Source location: Section 8.3

Commentary

A material model tells the shader how a surface responds to light. Diffuse material scatters light broadly; specular material creates view-dependent highlights; ambient terms approximate background illumination.

Material coefficients are not arbitrary decoration. They scale the contribution of each lighting term. If the specular coefficient is zero, the surface should not show a specular highlight under that model.

Mental model

Light asks what arrives; material answers how the surface reacts.

Check yourself

Which material parameter would you adjust to reduce shiny highlights?

08.4 Phong Illumination Model

Source location: Section 8.4

Commentary

The Phong illumination model combines ambient, diffuse, and specular terms. Diffuse lighting uses the angle between normal and light direction. Specular lighting also depends on the viewer because highlights move with the view direction.

The formula is less important than the decomposition. Ambient is a rough constant approximation. Diffuse is orientation-dependent matte reflection. Specular is view-dependent shininess. A correct answer should name the vectors and normalize them.

Mental model

Phong-style lighting is a sum of simple terms, each modeling a different visual effect.

Check yourself

Why does the specular term depend on the view direction?

08.5 Shading

Source location: Section 8.5

Commentary

Shading decides where the illumination calculation is evaluated and how results are interpolated. Flat shading uses one value per primitive. Gouraud shading evaluates at vertices and interpolates colors. Phong shading interpolates normals and evaluates lighting per fragment.

Per-fragment shading is more expensive but can represent highlights more accurately. This connects directly to the pipeline: interpolation during rasterization provides the data used by the fragment shader.

Mental model

The shading method decides what is computed at vertices and what is computed at fragments.

Check yourself

Why can Gouraud shading miss a small specular highlight?

End-of-lecture summary

If you remember only one thing from Lecture 08, remember this: This lecture explains how visible surface points receive color from local light, material, normal, and view information. It is about shading a point, not solving all global light transport.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.